Worksheet 7.2 — Equations of Planes

Grade 12 Calculus and Vectors (MCV4U) · Unit 7: Lines and Planes

The Big Ideas

A plane needs a point and a normal $\vec{n} = (A, B, C)$. Scalar: $Ax + By + Cz = D$ (the coefficients are the normal). Get a normal from two directions via the cross product.

- Scalar: $Ax + By + Cz = D$, normal (A, B, C).
- $A(x - x_0) + B(y - y_0) + C(z - z_0) = 0$.
- Normal from directions: $\vec{n} = \vec{u} \times \vec{v}$.

The Big Picture

Point and normal

A plane is fixed by a point and a **normal vector** perpendicular to it; the normal's components give the **scalar equation** $ax + by + cz + d = 0$.

Vector and parametric forms

Using two direction vectors lying in the plane, $\vec{r} = \vec{r}_0 + s\,\vec{u} + t\,\vec{v}$ describes every point on it.

Reading the normal

The coefficients (a, b, c) in the scalar equation are exactly the components of the normal vector.

Worked Examples

Example 1 — From point & normal

Plane through $(1, 2, 3)$ with normal $(2, 1, -1)$.

Solution. **Step 1 (apply $A(x - x_0) + B(y - y_0) + C(z - z_0) = 0$):** with $(A, B, C) = (2, 1, -1)$ and $(x_0, y_0, z_0) = (1, 2, 3)$: $2(x - 1) + 1(y - 2) - 1(z - 3) = 0$.

Step 2 (expand): $2x - 2 + y - 2 - z + 3 = 0$.

Step 3 (simplify and isolate the constant): $2x + y - z - 1 = 0 \Rightarrow 2x + y - z = 1$.

Conclusion: $2x + y - z = 1$.

Example 2 — Read the normal

Normal of $2x + y - z = 5$.

Solution. **Step 1 (recall that in $Ax+By+Cz = D$, the coefficients form the normal):** (A, B, C).

Conclusion: normal $= (2, 1, -1)$.

Example 3 — Point on the plane?

Is $(1, 1, 1)$ on $2x + y - z = 2$?

Solution. **Step 1 (substitute the point into the plane's equation):** $2(1) + 1 - 1$.

Step 2 (compare to the right side): $2 + 1 - 1 = 2$, matching the required value of 2.

Conclusion: yes, $(1, 1, 1)$ is on the plane.

Example 4 — Normal from directions

Plane contains $\vec{u} = (1, 0, 0), \vec{v} = (0, 1, 0)$. Find a normal.

Solution. **Step 1 (recall that a normal is perpendicular to both in-plane directions, found via the cross product):** $\vec{n} = \vec{u} \times \vec{v}$.

Step 2 (compute the cross product): $(0 \cdot 0 - 0 \cdot 1,\ 0 \cdot 0 - 1 \cdot 0,\ 1 \cdot 1 - 0 \cdot 0) = (0, 0, 1)$.

Conclusion: $\vec{n} = (0, 0, 1)$.

Example 5 — Through three points

Plane through $A(0, 0, 0), B(1, 0, 0), C(0, 1, 0)$.

Solution. **Step 1 (build two in-plane direction vectors from the points):** $\vec{AB} = (1, 0, 0)$, $\vec{AC} = (0, 1, 0)$.

Step 2 (find the normal via the cross product): $\vec{AB} \times \vec{AC} = (0, 0, 1)$.

Step 3 (write the equation using $A(0, 0, 0)$ as the point): $0(x - 0) + 0(y - 0) + 1(z - 0) = 0$.

Conclusion: $z = 0$.

Example 6 — From point & normal

Plane through $(0, 0, 0)$ with normal $(1, 2, 3)$.

Solution. **Step 1 (apply $A(x - x_0) + B(y - y_0) + C(z - z_0) = 0$ with the origin as the point):** $1(x - 0) + 2(y - 0) + 3(z - 0) = 0$.

Conclusion: $x + 2y + 3z = 0$.

Example 7 — Read the normal

Normal of $3x - y + 4z = 7$.

Solution. **Step 1 (read the coefficients of x, y, z directly):** $(3, -1, 4)$.

Conclusion: normal $= (3, -1, 4)$.

Example 8 — Point on the plane?

Is $(2, 0, 1)$ on $x + y + z = 3$?

Solution. **Step 1 (substitute the point):** $2 + 0 + 1$.

Step 2 (compare to 3): $2 + 0 + 1 = 3$, matches.

Conclusion: yes.

Example 9 — Vertical plane

Plane through $(2, 0, 0)$ with normal $(1, 0, 0)$.

Solution. **Step 1 (apply the point-normal formula):** $1(x - 2) + 0(y - 0) + 0(z - 0) = 0$.

Step 2 (simplify): $x - 2 = 0$.

Conclusion: $x = 2$.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Scalar equation through $(0, 0, 0)$ with normal $(1, 2, 3)$.

Question 2

Normal vector of $3x - y + 4z = 7$.

Question 3

Is $(2, 0, 1)$ on $x + y + z = 3$?

Question 4

Normal of the plane containing $(1, 0, 0)$ and $(0, 0, 1)$.

Question 5

Scalar equation through $(2, 0, 0)$ with normal $(1, 0, 0)$.

Question 6

Scalar equation through $(1, 1, 1)$ with normal $(1, 1, 1)$.

Question 7

Normal vector of $x - 2y + 5z = 0$.

Question 8

Is $(0, 3, 0)$ on $2x + y - z = 3$?

Question 9

Equation of the plane $z = 0$'s normal.

Question 10

Scalar equation through $(0, 0, 4)$ with normal $(0, 0, 1)$.

Question 11

Normal of the plane through $A(0, 0, 0), B(2, 0, 0), C(0, 3, 0)$.

Question 12

Is $(1, 2, 3)$ on $x + y + z = 6$?

Question 13 — Challenge

Find the scalar equation of the plane through $A(1, 0, 0), B(0, 1, 0), C(0, 0, 1)$.

Question Recap & Answer Key

Question Recap

1. Scalar equation through $(0,0,0)$ with normal $(1,2,3)$.
2. Normal vector of $3x - y + 4z = 7$.
3. Is $(2,0,1)$ on $x + y + z = 3$?
4. Normal of the plane containing $(1,0,0)$ and $(0,0,1)$.
5. Scalar equation through $(2,0,0)$ with normal $(1,0,0)$.
6. Scalar equation through $(1,1,1)$ with normal $(1,1,1)$.
7. Normal vector of $x - 2y + 5z = 0$.
8. Is $(0,3,0)$ on $2x + y - z = 3$?
9. Equation of the plane $z = 0$'s normal.
10. Scalar equation through $(0,0,4)$ with normal $(0,0,1)$.
11. Normal of the plane through $A(0,0,0), B(2,0,0), C(0,3,0)$.
12. Is $(1,2,3)$ on $x + y + z = 6$?
13. Find the scalar equation of the plane through $A(1,0,0), B(0,1,0), C(0,0,1)$.

✓ Answer Key

1. $x + 2y + 3z = 0$
2. $(3,-1,4)$
3. yes $(3 = 3)$
4. $(0,-1,0)$
5. $x = 2$
6. $x + y + z = 3$
7. $(1,-2,5)$
8. yes $(3 = 3)$
9. $(0,0,1)$
10. $z = 4$
11. $(0,0,6)$ or $(0,0,1)$
12. yes $(6 = 6)$
13. $\vec{n} = \vec{AB} \times \vec{AC} = (1,1,1)$; $x + y + z = 1$